

Revised architecture

Layer	Old	New
Backend	FastAPI	FastAPI (unchanged)
Database	PostgreSQL	PostgreSQL (or even SQLite, given your volume — one file, zero server admin)
Frontend	React + Tailwind	Unchanged
OCR	Google Cloud Vision (external API)	PaddleOCR (PP-OCRv4, mobile/lightweight model), running as a Python module inside FastAPI
Face verification	AWS Rekognition (external API)	InsightFace (buffalo_l or buffalo_s model), running locally — store a face embedding per student, compare via cosine similarity at pickup
Email	Microsoft Graph API	Unchanged
Scheduler	APScheduler	Unchanged